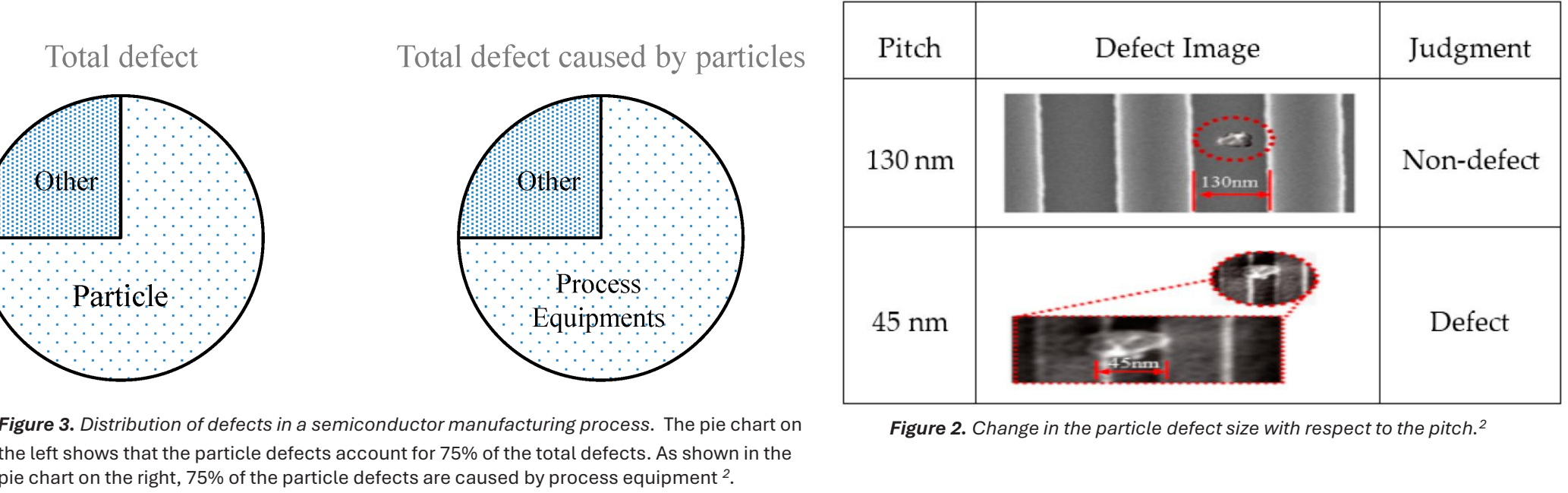


Real-Time Versatile Structured Light Imaging (rtVSLI)

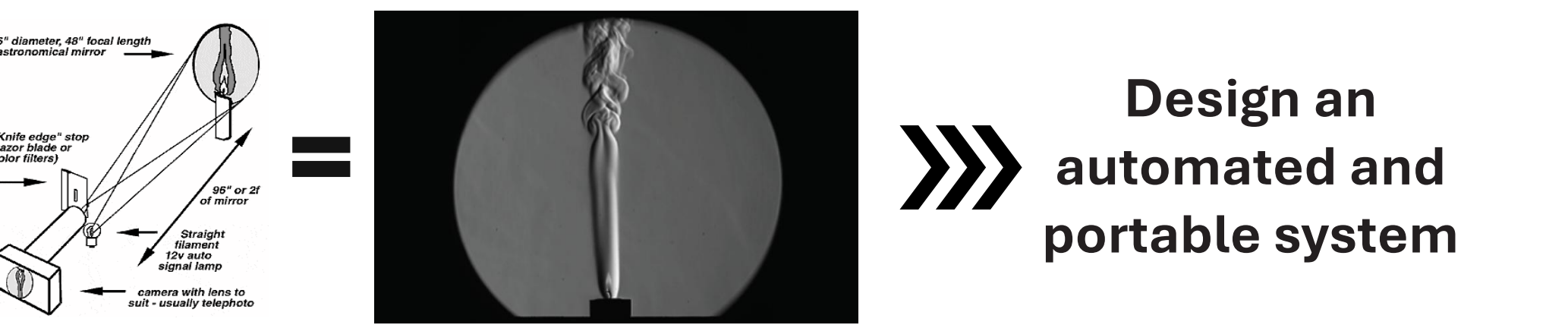
John Greenhalge, Rebekah Cross PhD, Battelle Memorial Institute

Problem

- Advancing product technology is outpacing environmental monitoring technology
- Contamination events cost companies hundreds of millions of dollars and represent significant safety concerns in high containment environments.
 - Current monitoring procedures generally provide low resolution data, are reactive instead of proactive, and can amount to hot-spot measurements.
 - Identifying contamination sources often requires costly investigations, downtime, and repeat testing that adds no direct value to production.
 - Advancing product technology necessitates advanced environmental monitoring technology yet current monitoring methods are significantly behind.



Solution - rtVSLI



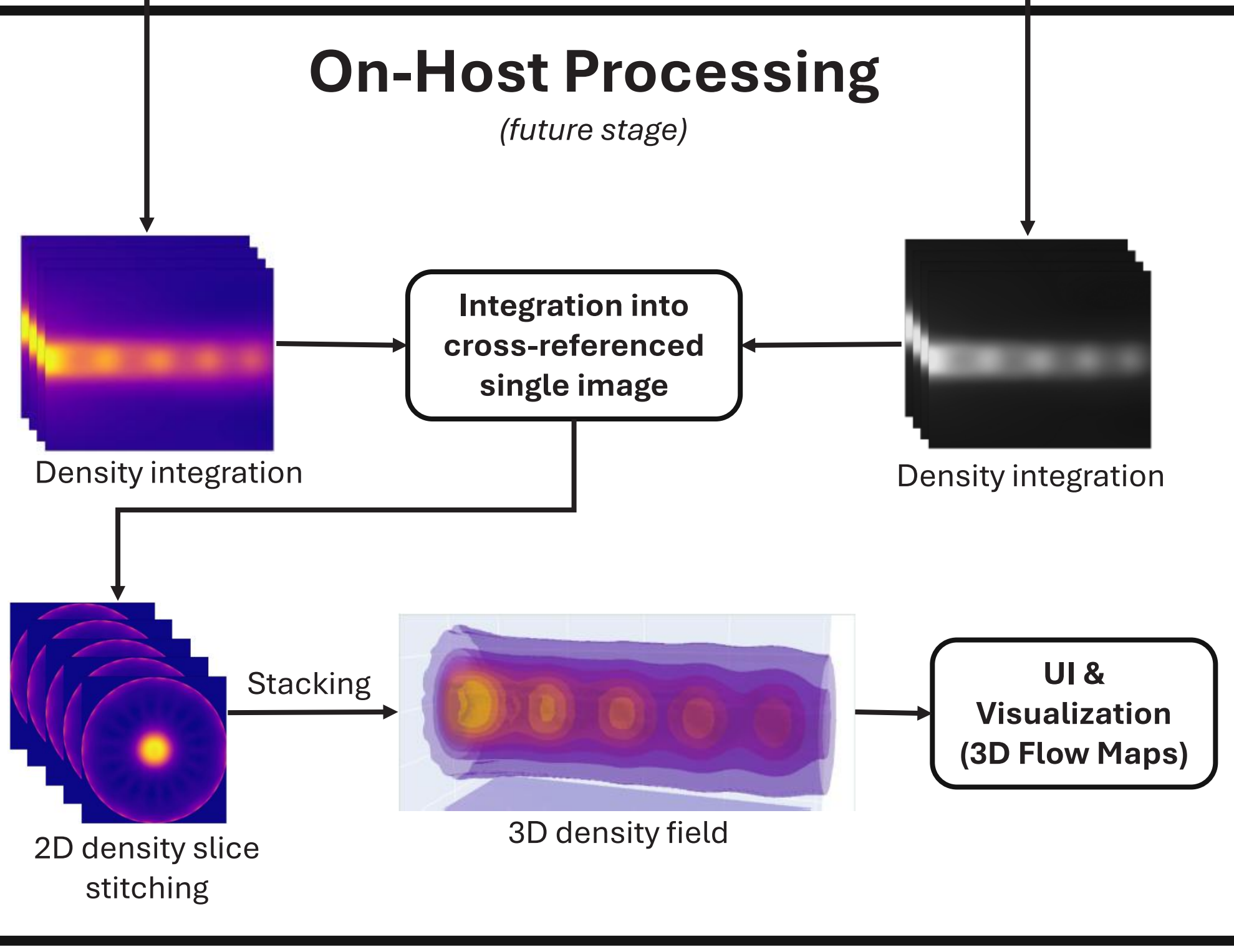
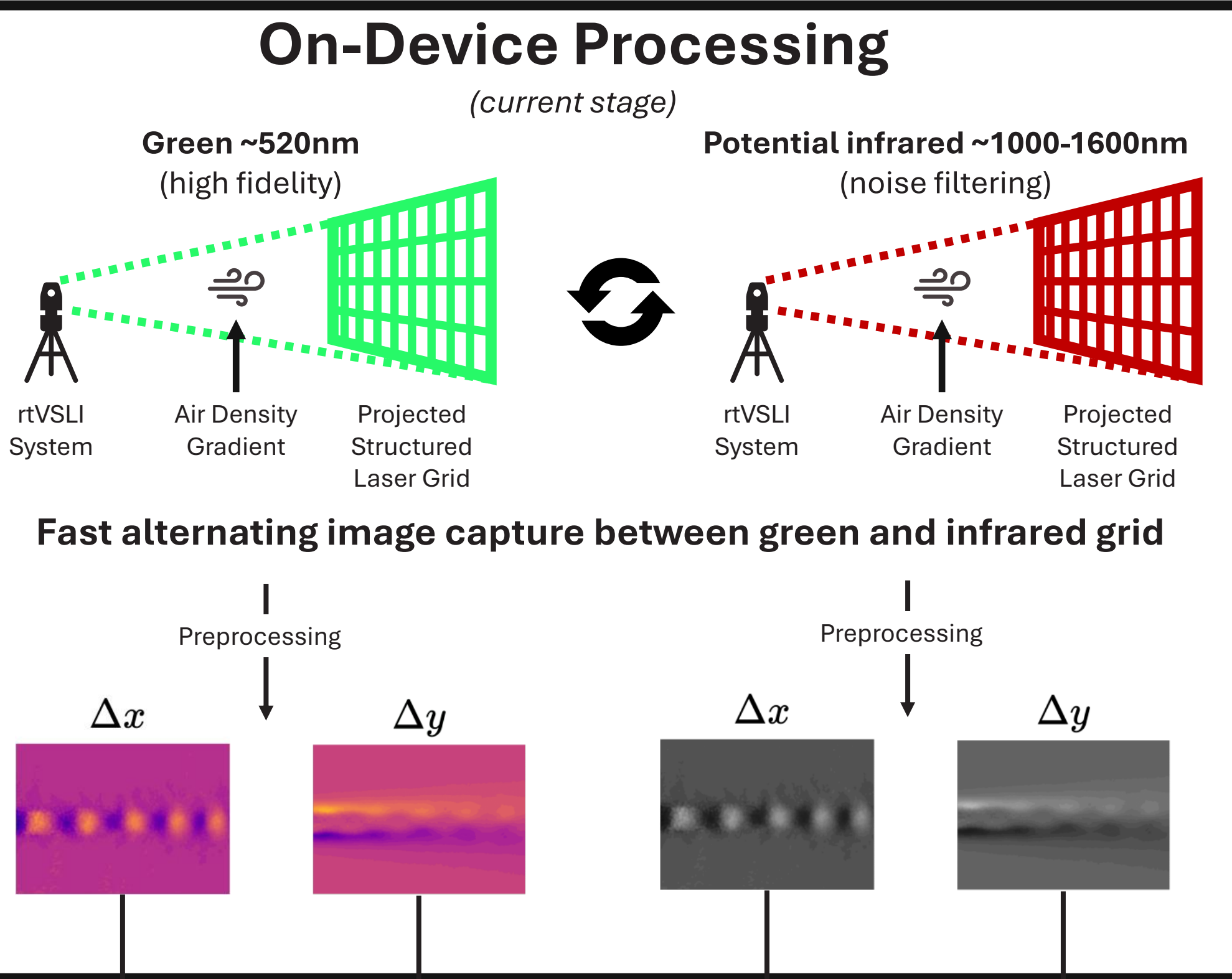
Give industries an intuitive and continuous view of the environments that protect their products by turning airflow into actionable insight for contamination prevention

- Project laser-grid background onto the environment and measure how the light's path bends as it moves through the air
- Using a slide projector and camera array, the laser grid can be moved and scaled to adjust for distances
- Projected background can be perturbed by a rotating mirror, allowing evaluation with reference-free shadowgraph or BOS methods¹

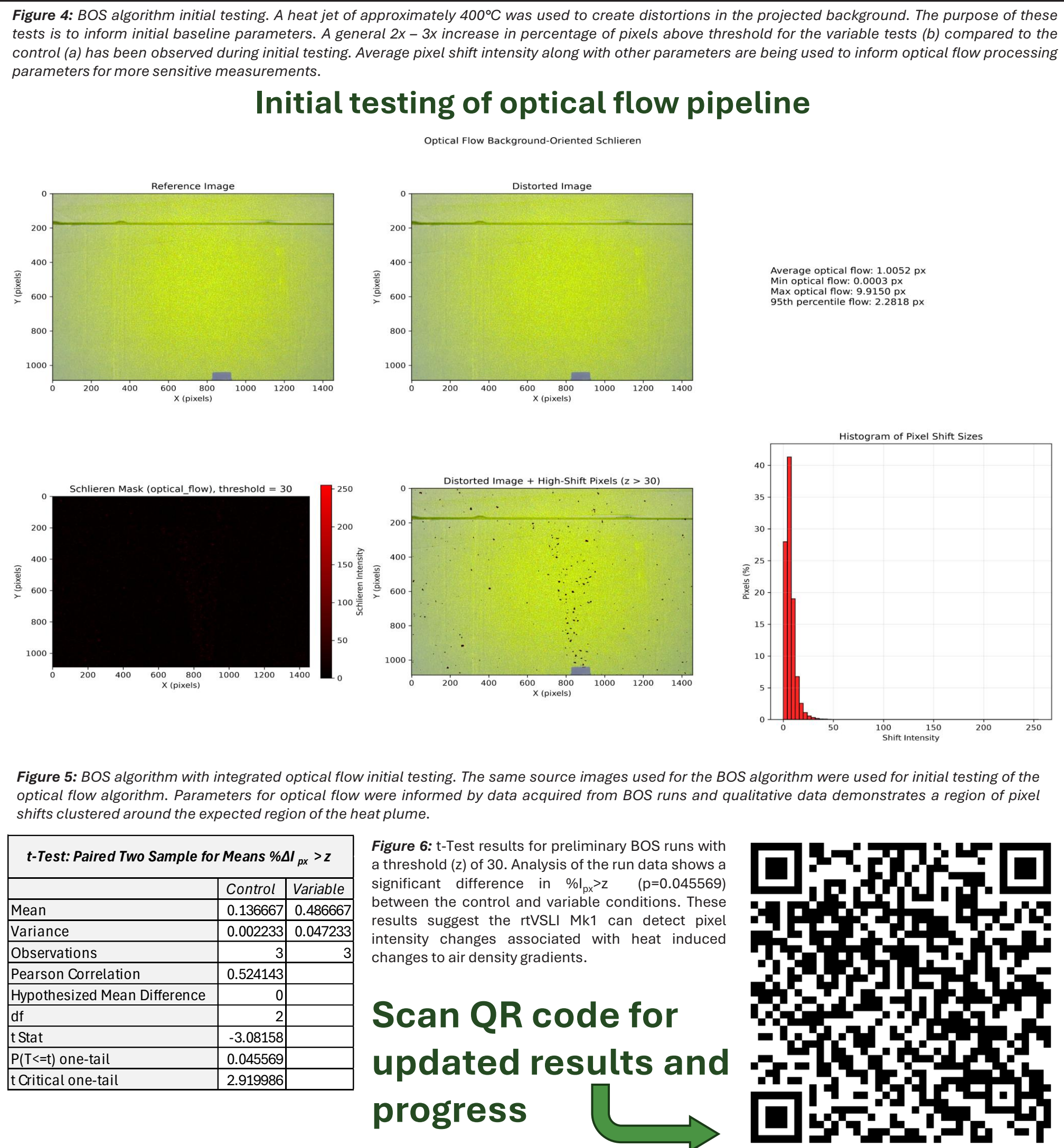
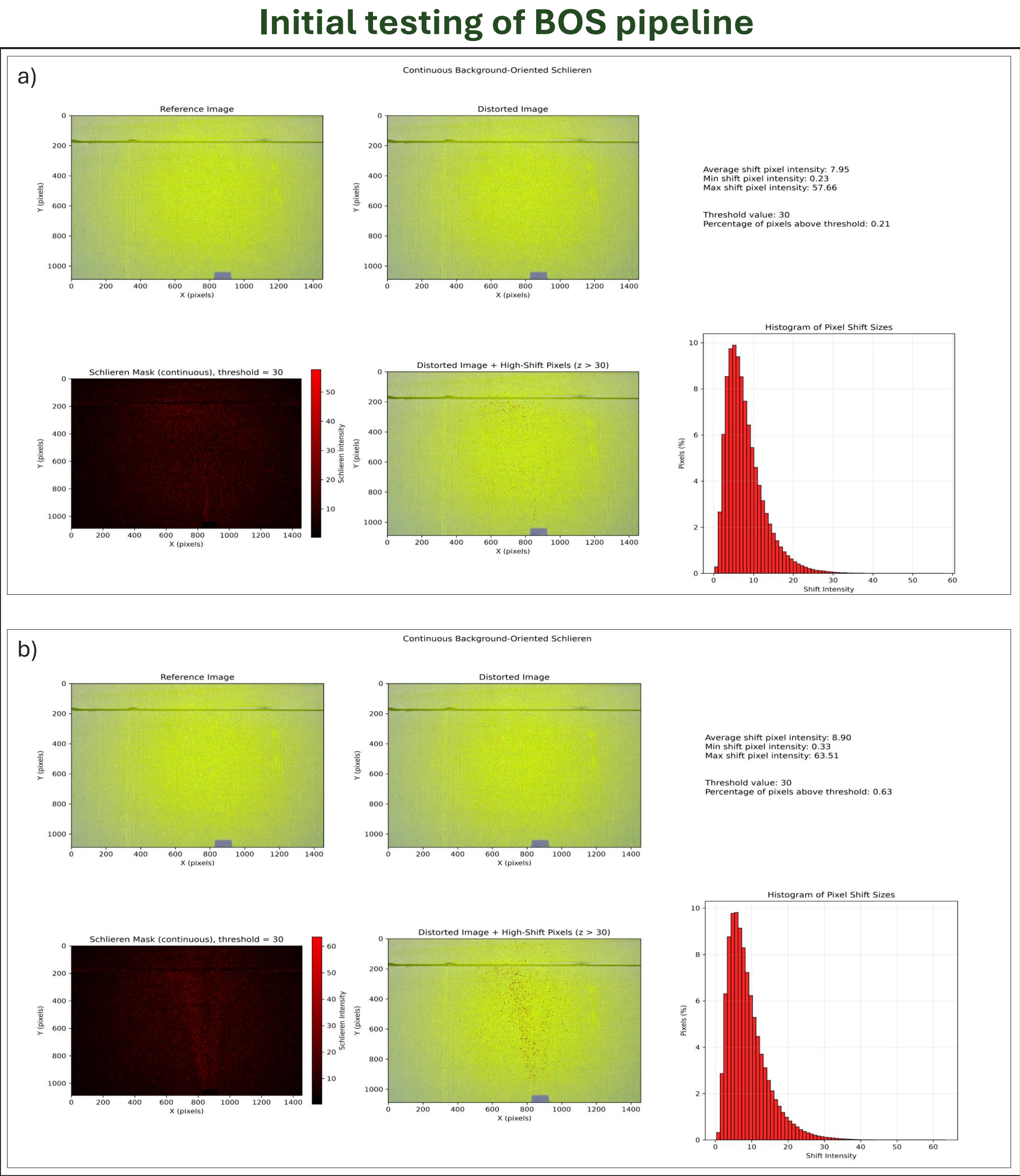


Approach^{1,3}

- Project a structured laser grid background onto the environment
- From the same device, image distortions in background caused by air density gradients
- Use of two wavelengths for high fidelity and noise control, if necessary
- Controlled perturbations of projected background to separate the variable and establish a baseline
- Cross reference output from both wavelengths for a synthesized differential measurement and visual.



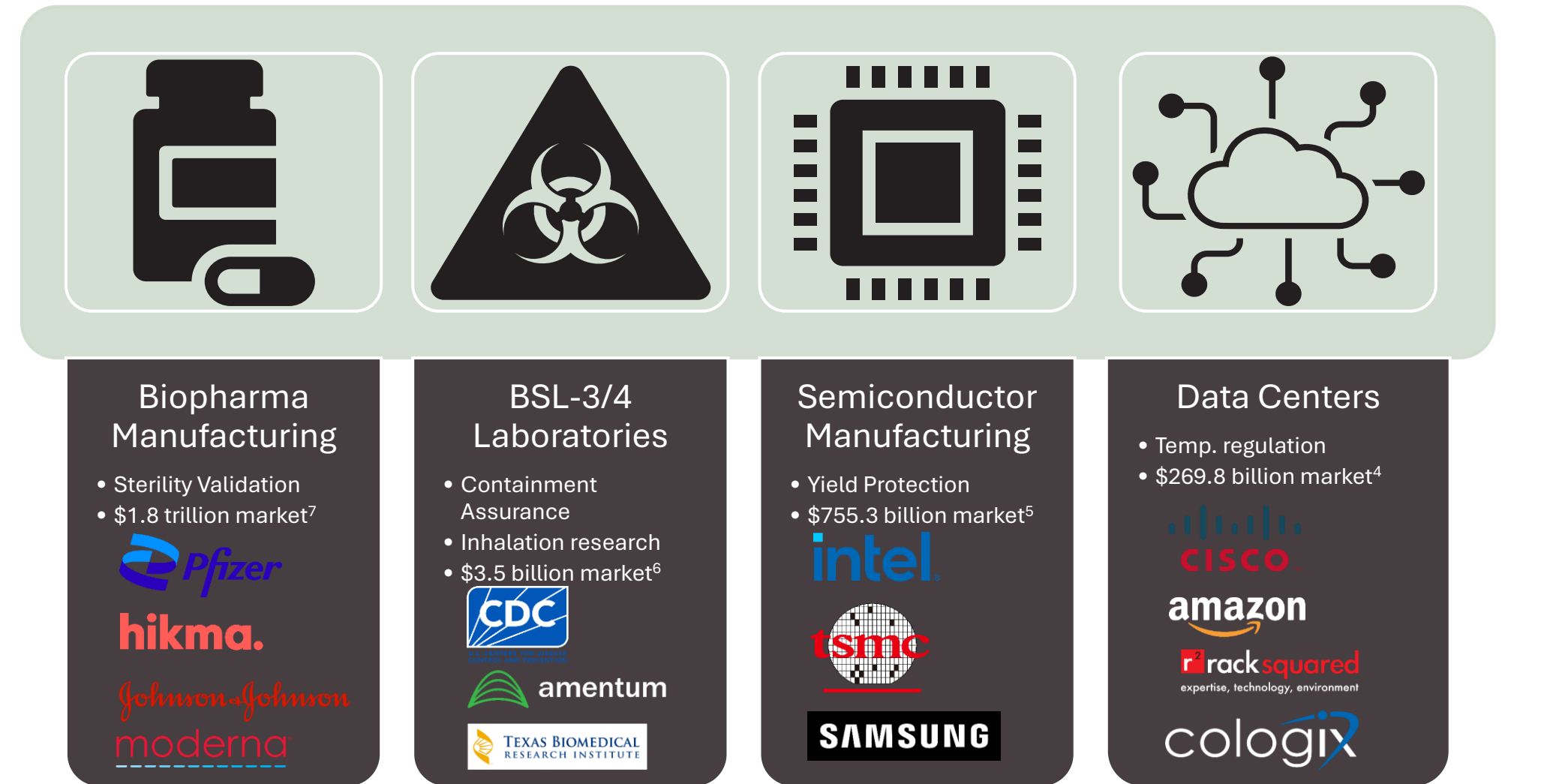
Results



Next Steps

- De-risking of current prototype architecture and scripts
- Refinement of BOS and optical flow processing parameters
- Video processing
- Institute reference-free differential imaging
- Data summary and report for Mk1 prototype
- Internal or external funding for Mk2 to begin integrating parallax and dual-wavelength differential imaging

Market Assessment



- Based on most comparable systems, final product cost (if sold direct to consumer) would be ~\$200k-\$500k^{8,9}
- Potential direct to consumer or provided service business model
- Average cost of one recall in pharma or semiconductor manufacturing is between \$50M-\$600M just for product cost. This doesn't include indirect costs like litigation, settlements, or brand damage^{10,11}

Market Estimates	Top-Down Assessment	Bottom-Up Assessment
Total Accessible Market (Global)	~\$4.1B per year	~\$350 - 400M per year
Total Accessible Market (U.S.)	~\$1.0 - 1.3B per year	~\$100 - 110M per year
Serviceable Available Market (Global)	~\$300 - 400M per year	~\$120 - 130M per year
Serviceable Available Market (U.S.)	~\$150 - 170M per year	~\$30 - 35M per year

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